

Paavani Karuturi

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EXPERIENCE

AI Engineer

Jun 2023 - Aug 2023

Fractal Analytics

Mumbai, India

- Shipped 12 production features across **Python**, **TypeScript**, and **Java** for a platform reaching millions of users, carrying requirements through object-oriented design, implementation, testing, and release.
- Reworked **PostgreSQL** indexes and introduced **Redis** caching on hot read paths, eliminating 60% of redundant computation and reducing query latency by roughly one-third.
- Used AI coding assistants for test scaffolding and debugging, integrating **Jest** and **Cypress** into **GitHub Actions** to raise coverage 40% and reduce regressions 30%.

Full Stack Engineer

Dec 2022 - May 2023

Darwinbox

Hyderabad, India

- Built and optimized **FastAPI** and REST services backed by **MySQL** indexes and query plans, reducing p95 dashboard latency 30% for workflows serving 10K+ users.
- Applied request validation, caching, and data-access improvements to reduce redundant traffic 20%, while debugging production edge cases and documenting API behavior for consuming teams.
- Automated **Docker**-based delivery through **GitHub Actions**, increasing deployment frequency 40% while standardizing repeatable testing and release procedures.

Software Engineer

Jun 2021 - Dec 2022

Rivach

Hyderabad, India

- Built **Java** REST APIs and **MySQL** data models for backend services serving 25K daily users, resolving concurrency failures under high-frequency writes while preserving data integrity.
- Analyzed query execution plans and tuned database access patterns, reducing p95 latency roughly 30% while maintaining correctness under concurrent load.
- Collaborated through code reviews and production troubleshooting, converting ambiguous failures into scoped fixes, test cases, and documented handoffs for reliable delivery.

EDUCATION

San Jose State University

San Jose, CA

Master of Science in Software Engineering, GPA: 3.8/4.0

Aug 2024 - May 2026

SKILLS

Languages: Python, Java, C++, Go, TypeScript, JavaScript, SQL

Systems: Distributed systems, concurrency, parallelism, data structures, algorithms, object-oriented design, Unix/Linux, networking

Cloud and Infrastructure: Kubernetes, Docker, AWS, GCP, Kafka, GitHub Actions, CI/CD, Prometheus, Grafana, OpenTelemetry

Data Platforms: PostgreSQL, MySQL, Cassandra, MongoDB, Redis, indexing, caching, data pipelines, NoSQL systems

AI and ML: CUDA, vLLM, Triton, LangGraph, LangChain, MCP, RAG, LLM evaluation, prompt engineering, AI coding assistants

Engineering: Unit testing, integration testing, debugging, troubleshooting, code reviews, structured logging, performance optimization, SDLC

SIDE PROJECTS

AegisFlow | *Go, Docker, Kubernetes, OpenTelemetry, Prometheus, Redis* | [GitHub](#)

- Built an open-source AI gateway in **Go** with circuit-breaker and self-healing failover paths, sustaining 58K requests per second under high-throughput traffic.
- Implemented **Redis**-backed rate limiting and containerized deployment with **Docker** and **Kubernetes**, designing explicit control points for resilient multi-service operation.
- Instrumented request paths with **OpenTelemetry** and **Prometheus**, then added 48+ tests to validate failure recovery, rate-limit behavior, and maintainability.

CERTIFICATIONS & ACHIEVEMENTS

AWS Certified Solutions Architect - Associate | Google Cloud Computing Foundations | AWS Academy Graduate

Runner-up - Stanford LLM x Law Hackathon (2025) | Winner - Cloud Native Hackathon (2022) | Google Cloud: Generative AI with Vertex AI